

KITABU QUEST

S3

MATHEMATICS



Cover scene

learners solving mathematics with charts and measuring tools with a mountain gorilla mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

3

Title Page

KITABU QUEST S3 Mathematics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S3 learners study Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use objects, drawings, number sentences, and worked examples before independent practice.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Problem sets
2. Number bases
3. Algebraic fractions
4. Simultaneous linear equations and inequalities
5. Quadratic equations
6. Linear and quadratic functions
7. Compound interest, reverse percentage, and compound proportional change
8. Right angle triangles
9. Circle theorem
10. Collinear points and orthogonal vectors
11. Enlargement and similarity in 2D
12. Inverse and composite transformations in 2D

As you finish each topic, return to the success criteria and correct one answer before moving on.

Problem sets: Lesson Opener

Learning focus: Problem sets

Country link: a Rwandan school, market, farming, building, transport, or data problem for Problem sets.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve problem sets
- express a mathematical problem set using a Venn diagram
- represent a mathematical problem using a Venn diagram
- use Venn diagram to represent a mathematical problem set
- interpret, model, and solve a mathematical problem set
- develop clear, logical and coherent thinking life problems involving sets

Inquiry questions:

- How can problem sets help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Problem sets: Learn

Problem sets teaches how heavy objects are.

Core idea:

- Mass is measured with units such as grams and kilograms.
- A balance or scale helps compare mass fairly.
- Convert units before adding or comparing if the units are different.

Key words:

- Problem sets
- Problem
- sets
- method
- model
- unit
- answer
- check

Problem sets: Worked Example

Worked example for Problem sets: A bag of beans has a mass of 3 kg. A small packet has a mass of 500 g. What is the total mass in grams?

1. Convert 3 kg to grams: $3 \text{ kg} = 3000 \text{ g}$.
2. Add $3000 \text{ g} + 500 \text{ g} = 3500 \text{ g}$.

Answer: 3500 g.

Problem sets: Vocabulary And Study Skill

Use these words and study moves before you practise Problem sets. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Problem sets
- Problem
- sets
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Problem sets without a clear example, method, or evidence.

Problem sets: Guided Activity

Guided activity for Problem sets:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Problem sets: Practice And Correction

Practice for Problem sets:

Main outcome: solve problem sets.

1. Name two objects measured in grams and two measured in kilograms.
2. Convert 2 kg to grams.
3. Compare 1500 g and 2 kg. Which is heavier?

Answer check:

Answer: 2 kg = 2000 g, so 2 kg is heavier than 1500 g.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to problem sets and solve it.

Problem sets: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Problem sets.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Problem sets: Outcome Check

Outcome check for Problem sets: Solve problem sets.

1. Convert 4 kg to grams.
2. Compare 2500 g and 3 kg.
3. Write the heavier mass with a reason.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Problem sets: Outcome Check

Outcome check for Problem sets: Express a mathematical problem set using a Venn diagram.

1. Convert 4 kg to grams.
2. Compare 2500 g and 3 kg.
3. Write the heavier mass with a reason.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Number bases: Lesson Opener

Learning focus: Number bases

Country link: a Rwandan school, market, farming, building, transport, or data problem for Number bases.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- represent numbers in different number bases and solve related problems
- list the digits used in a given base
- convert numbers from base ten to any other base and vice versa
- carry out addition, subtraction, multiplication, and division on numbers bases
- solve equations involving bases
- develop clear, logical and coherent thinking while solving problem sets

Inquiry questions:

- How can number bases help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Number bases: Learn

Number bases teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:

- Number bases
- Number
- bases
- method
- model
- unit
- answer
- check

Number bases: Worked Example

Worked example for Number bases: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:

1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Number bases: Vocabulary And Study Skill

Use these words and study moves before you practise Number bases. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Number bases
- Number
- bases
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Number bases without a clear example, method, or evidence.

Number bases: Guided Activity

Guided activity for Number bases:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Number bases: Practice And Correction

Practice for Number bases:

Main outcome: represent numbers in different number bases and solve related problems.

1. Solve one number bases question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to number bases and solve it.

Number bases: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Number bases.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Number bases: Outcome Check

Outcome check for Number bases: Represent numbers in different number bases and solve related problems.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Number bases: Outcome Check

Outcome check for Number bases: List the digits used in a given base.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Algebraic fractions: Lesson Opener

Learning focus: Algebraic fractions

Country link: a Rwandan school, market, farming, building, transport, or data problem for Algebraic fractions.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- perform operations on rational expressions and use them in different situations
- define an algebraic fraction
- state the restriction on the variable in algebraic fraction
- recognise the rules applied to addition, subtraction, multiplication, division and simplification of algebraic fractions
- perform operations on algebraic fractions
- solve rational equations with linear denominators

Inquiry questions:

- How can algebraic fractions help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Algebraic fractions: Learn

Algebraic fractions teaches how parts of a whole are named and combined.

Core idea:

- The denominator tells how many equal parts make one whole.
- The numerator tells how many of those parts are being used.
- Fractions with the same denominator can be added or subtracted by working with the numerators.
- The denominator stays the same because the size of the parts has not changed.

Key words:

- Algebraic fractions
- Algebraic
- fractions
- method
- model
- unit
- answer
- check

Algebraic fractions: Worked Example

Worked example for Algebraic fractions: A learner eats $\frac{2}{8}$ of a pineapple in the morning and $\frac{3}{8}$ in the afternoon. What fraction is eaten?

1. The denominators are the same, so the parts are equal-sized eighths.
2. Add the numerators: $2 + 3 = 5$.
3. Keep the denominator: 8.

Answer: $\frac{5}{8}$ of the pineapple was eaten.

Algebraic fractions: Vocabulary And Study Skill

Use these words and study moves before you practise Algebraic fractions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Algebraic fractions
- Algebraic
- fractions
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Algebraic fractions without a clear example, method, or evidence.

Algebraic fractions: Guided Activity

Guided activity for Algebraic fractions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Algebraic fractions: Practice And Correction

Practice for Algebraic fractions:

Main outcome: perform operations on rational expressions and use them in different situations.

1. Draw one whole divided into 8 equal parts. Shade $\frac{3}{8}$.
2. Calculate $\frac{2}{7} + \frac{4}{7}$ and explain why the denominator stays 7.
3. Write one story problem using fractions with the same denominator, then solve it.

Answer check:

Answers: $\frac{2}{7} + \frac{4}{7} = \frac{6}{7}$. The denominator stays 7 because the equal parts are sevenths.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to algebraic fractions and solve it.

Algebraic fractions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Algebraic fractions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Algebraic fractions: Outcome Check

Outcome check for Algebraic fractions: Perform operations on rational expressions and use them in different situations.

1. Draw $\frac{5}{8}$ on a fraction strip.
2. Calculate $\frac{3}{9} + \frac{4}{9}$.
3. Explain why the denominator does not change.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Algebraic fractions: Outcome Check

Outcome check for Algebraic fractions: Define an algebraic fraction.

1. Draw $\frac{5}{8}$ on a fraction strip.
2. Calculate $\frac{3}{9} + \frac{4}{9}$.
3. Explain why the denominator does not change.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Simultaneous linear equations and inequalities: Lesson Opener

Learning focus: Simultaneous linear equations and inequalities

Country link: a Rwandan school, market, farming, building, transport, or data problem for Simultaneous linear equations and inequalities.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve word problems involving simultaneous linear equations and inequalities
- define simultaneous linear inequalities in two unknowns
- give examples of simultaneous linear inequalities in two unknowns
- show solution set of simultaneous linear equations and inequalities in two unknowns given their graphs
- solve graphically simultaneous linear equations and inequalities in two unknowns
- interpret graphical solutions of simultaneous linear equations and inequalities in two unknowns

Inquiry questions:

- How can simultaneous linear equations and inequalities help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Simultaneous linear equations and inequalities: Learn

Simultaneous linear equations and inequalities teaches how lines and angles describe shapes and directions.

Core idea:

- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:

- Simultaneous linear equations and inequalities
- Simultaneous
- linear
- equations
- inequalities
- method
- model
- unit

Simultaneous linear equations and inequalities: Worked Example

Worked example for Simultaneous linear equations and inequalities: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Simultaneous linear equations and inequalities: Vocabulary And Study Skill

Use these words and study moves before you practise Simultaneous linear equations and inequalities. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Simultaneous linear equations and inequalities
- Simultaneous
- linear
- equations
- inequalities
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Simultaneous linear equations and inequalities without a clear example, method, or evidence.

Simultaneous linear equations and inequalities: Guided Activity

Guided activity for Simultaneous linear equations and inequalities:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Simultaneous linear equations and inequalities: Practice And Correction

Practice for Simultaneous linear equations and inequalities:

Main outcome: solve word problems involving simultaneous linear equations and inequalities.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:

Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to simultaneous linear equations and inequalities and solve it.

Simultaneous linear equations and inequalities: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Simultaneous linear equations and inequalities.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Simultaneous linear equations and inequalities: Outcome Check

Outcome check for Simultaneous linear equations and inequalities: Solve word problems involving simultaneous linear equations and inequalities.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Simultaneous linear equations and inequalities: Outcome Check

Outcome check for Simultaneous linear equations and inequalities: Define simultaneous linear inequalities in two unknowns.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Quadratic equations: Lesson Opener

Learning focus: Quadratic equations

Country link: a Rwandan school, market, farming, building, transport, or data problem for Quadratic equations.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve quadratic equations
- define quadratic equations
- state methods used to solve a quadratic equation
- solve quadratic equations
- model and solve problems involving quadratic equations
- solve equations reducible to quadratic

Inquiry questions:

- How can quadratic equations help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Quadratic equations: Learn

Quadratic equations teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:

- Quadratic equations
- Quadratic
- equations
- method
- model
- unit
- answer
- check

Quadratic equations: Worked Example

Worked example for Quadratic equations: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:

1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Quadratic equations: Vocabulary And Study Skill

Use these words and study moves before you practise Quadratic equations. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Quadratic equations
- Quadratic
- equations
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Quadratic equations without a clear example, method, or evidence.

Quadratic equations: Guided Activity

Guided activity for Quadratic equations:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Quadratic equations: Practice And Correction

Practice for Quadratic equations:

Main outcome: solve quadratic equations.

1. Solve one quadratic equations question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to quadratic equations and solve it.

Quadratic equations: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Quadratic equations.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Quadratic equations: Outcome Check

Outcome check for Quadratic equations: Solve quadratic equations.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Quadratic equations: Outcome Check

Outcome check for Quadratic equations: Define quadratic equations.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Linear and quadratic functions: Lesson Opener

Learning focus: Linear and quadratic functions

Country link: a Rwandan school, market, farming, building, transport, or data problem for Linear and quadratic functions.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve problems involving linear or quadratic functions and interpret the graphs of quadratic functions
- define a Cartesian equation of a straight line
- define quadratic function
- list the characteristics of linear or quadratic function
- differentiate linear from quadratic functions
- determine Cartesian equations of straight lines, coordinates of vertices, the equation of axis of symmetry

Inquiry questions:

- How can linear and quadratic functions help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Linear and quadratic functions: Learn

Linear and quadratic functions teaches how lines and angles describe shapes and directions.

Core idea:

- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:

- Linear and quadratic functions
- Linear
- quadratic
- functions
- method
- model
- unit
- answer

Linear and quadratic functions: Worked Example

Worked example for Linear and quadratic functions: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Linear and quadratic functions: Vocabulary And Study Skill

Use these words and study moves before you practise Linear and quadratic functions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Linear and quadratic functions
- Linear
- quadratic
- functions
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Linear and quadratic functions without a clear example, method, or evidence.

Linear and quadratic functions: Guided Activity

Guided activity for Linear and quadratic functions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Linear and quadratic functions: Practice And Correction

Practice for Linear and quadratic functions:

Main outcome: solve problems involving linear or quadratics functions and interpret the graphs of quadratic functions.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:

Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to linear and quadratic functions and solve it.

Linear and quadratic functions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Linear and quadratic functions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Linear and quadratic functions: Outcome Check

Outcome check for Linear and quadratic functions: Solve problems involving linear or quadratic functions and interpret the graphs of quadratic functions.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Linear and quadratic functions: Outcome Check

Outcome check for Linear and quadratic functions: Define a Cartesian equation of a straight line.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Compound interest, reverse percentage, and compound proportional change: Lesson Opener

Learning focus: Compound interest, reverse percentage, and compound proportional change

Country link: a Rwandan school, market, farming, building, transport, or data problem for Compound interest, reverse percentage, and compound proportional change.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve problems involving compound interest, reverse percentage and proportional change using multipliers
- define compound interest, reverse percentage, compound proportional change, and continued proportional
- find reverse percentages in a given mathematical problem
- determine a compound interest in a given mathematical problem
- simplify ratio in their simplest form
- solve problems involving reverse percentages and compound interest

Inquiry questions:

- How can compound interest, reverse percentage, and compound proportional change help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Compound interest, reverse percentage, and compound proportional change: Learn

Compound interest, reverse percentage, and compound proportional change teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Compound interest, reverse percentage, and compound proportional change
- Compound
- interest
- reverse
- percentage
- compound
- proportional
- change

Compound interest, reverse percentage, and compound proportional change: Worked Example

Worked example for Compound interest, reverse percentage, and compound proportional change: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Compound interest, reverse percentage, and compound proportional change: Vocabulary And Study Skill

Use these words and study moves before you practise Compound interest, reverse percentage, and compound proportional change. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Compound interest, reverse percentage, and compound proportional change
- Compound
- interest
- reverse
- percentage
- compound
- proportional
- change

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Compound interest, reverse percentage, and compound proportional change without a clear example, method, or evidence.

Compound interest, reverse percentage, and compound proportional change: Guided Activity

Guided activity for Compound interest, reverse percentage, and compound proportional change:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Compound interest, reverse percentage, and compound proportional change: Practice And Correction

Practice for Compound interest, reverse percentage, and compound proportional change:

Main outcome: solve problems involving compound interest, reverse percentage and proportional change using multipliers.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to compound interest, reverse percentage, and compound proportional change and solve it.

Compound interest, reverse percentage, and compound proportional change: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Compound interest, reverse percentage, and compound proportional change.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Compound interest, reverse percentage, and compound proportional change: Outcome Check

Outcome check for Compound interest, reverse percentage, and compound proportional change:
Solve problems involving compound interest, reverse percentage and proportional change using multipliers.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Compound interest, reverse percentage, and compound proportional change: Outcome Check

Outcome check for Compound interest, reverse percentage, and compound proportional change:
Define compound interest, reverse percentage, compound proportional change, and continued proportional.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Right angle triangles: Lesson Opener

Learning focus: Right angle triangles

Country link: a Rwandan school, market, farming, building, transport, or data problem for Right angle triangles.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- find lengths of sides and angles in right angle triangles using trigonometric ratios
- give and define the elements of a right angle triangle
- show relationship between the elements of a right angled triangle
- use Pythagoras' theorem to find the relationship between the elements of a right angle triangle
- solve problems about right angle triangle using the properties of elements of a right angle triangle and Pythagoras' theorem
- appreciate the importance of right angle triangles in various situations

Inquiry questions:

- How can right angle triangles help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Right angle triangles: Learn

Right angle triangles teaches properties of two-dimensional shapes.

Core idea:

- Shapes can be classified by sides, corners, lines of symmetry, and angles.
- A labelled sketch helps you compare shapes accurately.
- Use correct names such as triangle, rectangle, square, and circle.

Key words:

- Right angle triangles
- Right
- angle
- triangles
- method
- model
- unit

Right angle triangles: Worked Example

Worked example for Right angle triangles: Classify a shape with 4 equal sides and 4 right angles.

1. Four sides means it is a quadrilateral.
2. Equal sides and right angles match a square.
3. Check the name and properties.

Answer: The shape is a square.

Right angle triangles: Vocabulary And Study Skill

Use these words and study moves before you practise Right angle triangles. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Right angle triangles
- Right
- angle
- triangles
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Right angle triangles without a clear example, method, or evidence.

Right angle triangles: Guided Activity

Guided activity for Right angle triangles:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Right angle triangles: Practice And Correction

Practice for Right angle triangles:

Main outcome: find lengths of sides and angles in right angle triangles using trigonometric ratios.

1. Draw a triangle, rectangle, and square.
2. Write two properties for each shape.
3. Circle the shape that has four equal sides and four right angles.

Answer check:

Answer: the square has four equal sides and four right angles.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to right angle triangles and solve it.

Right angle triangles: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Right angle triangles.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Right angle triangles: Outcome Check

Outcome check for Right angle triangles: Find lengths of sides and angles in right angle triangles using trigonometric ratios.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Right angle triangles: Outcome Check

Outcome check for Right angle triangles: Give and define the elements of a right angle triangle.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Circle theorem: Lesson Opener

Learning focus: Circle theorem

Country link: a Rwandan school, market, farming, building, transport, or data problem for Circle theorem.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- construct mathematical arguments about circles and disks and use circle theorem to solve related problems
- recognise and identify the elements of a circle
- identify angle properties in a circle
- find the length of elements of a circle
- calculate the area of disk and its sector
- use the angle properties of lines in circles to solve problems

Inquiry questions:

- How can circle theorem help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Circle theorem: Learn

Circle theorem teaches the parts and measurements of circles.

Core idea:

- The centre is the fixed middle point of a circle.
- A radius goes from the centre to the circle.
- A diameter passes through the centre and is twice the radius.
- Circumference is the distance around the circle.
- A labelled diagram prevents mixing up radius and diameter.

Key words:

- Circle theorem
- Circle
- theorem
- method
- model
- unit
- answer
- check

Circle theorem: Worked Example

Worked example for Circle theorem: A circle has radius 7 cm. Find its diameter.

1. Diameter = 2 x radius.
2. Radius = 7 cm.
3. $2 \times 7 \text{ cm} = 14 \text{ cm}$.
4. Check the diagram: the diameter must pass through the centre.

Answer: The diameter is 14 cm.

Circle theorem: Vocabulary And Study Skill

Use these words and study moves before you practise Circle theorem. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Circle theorem
- Circle
- theorem
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Circle theorem without a clear example, method, or evidence.

Circle theorem: Guided Activity

Guided activity for Circle theorem:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Circle theorem: Practice And Correction

Practice for Circle theorem:

Main outcome: construct mathematical arguments about circles and disks and use circle theorem to solve related problems.

1. Draw a circle and label the centre, radius, and diameter.
2. If the radius is 6 cm, find the diameter.
3. Name one circular object at school or home and describe its radius or diameter.

Answer check:

Answer: if radius = 6 cm, diameter = 12 cm. A correct diagram shows the radius from centre to circle and diameter through the centre.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to circle theorem and solve it.

Circle theorem: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Circle theorem.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Circle theorem: Outcome Check

Outcome check for Circle theorem: Construct mathematical arguments about circles and disks and use circle theorem to solve related problems.

1. Draw and label a circle centre, radius, and diameter.
2. If radius is 4 cm, find diameter.
3. Explain why diameter is twice the radius.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Circle theorem: Outcome Check

Outcome check for Circle theorem: Recognise and identify the elements of a circle.

1. Draw and label a circle centre, radius, and diameter.
2. If radius is 4 cm, find diameter.
3. Explain why diameter is twice the radius.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Collinear points and orthogonal vectors: Lesson Opener

Learning focus: Collinear points and orthogonal vectors

Country link: a Rwandan school, market, farming, building, transport, or data problem for Collinear points and orthogonal vectors.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- apply properties of collinearity and orthogonality to solve problems involving vectors
- state the conditions and properties of collinearity and orthogonality
- use definition and properties to show whether three given points are collinear or not and whether two vectors are orthogonal or not
- appreciate the use of properties of collinearity and orthogonality to solve problems about vectors in two dimensions
- show patience, mutual respect, tolerance, and curiosity in solving and discussing problems involving vectors in two dimensions
- conditions for points to be collinear and vectors to be orthogonal

Inquiry questions:

- How can collinear points and orthogonal vectors help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Collinear points and orthogonal vectors: Learn

Collinear points and orthogonal vectors teaches how lines and angles describe shapes and directions.

Core idea:

- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:

- Collinear points and orthogonal vectors
- Collinear
- points
- orthogonal
- vectors
- method
- model
- unit

Collinear points and orthogonal vectors: Worked Example

Worked example for Collinear points and orthogonal vectors: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Collinear points and orthogonal vectors: Vocabulary And Study Skill

Use these words and study moves before you practise Collinear points and orthogonal vectors. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Collinear points and orthogonal vectors
- Collinear
- points
- orthogonal
- vectors
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Collinear points and orthogonal vectors without a clear example, method, or evidence.

Collinear points and orthogonal vectors: Guided Activity

Guided activity for Collinear points and orthogonal vectors:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Collinear points and orthogonal vectors: Practice And Correction

Practice for Collinear points and orthogonal vectors:

Main outcome: apply properties of collinearity and orthogonality to solve problems involving vectors.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:

Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to collinear points and orthogonal vectors and solve it.

Collinear points and orthogonal vectors: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Collinear points and orthogonal vectors.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Collinear points and orthogonal vectors: Outcome Check

Outcome check for Collinear points and orthogonal vectors: Apply properties of collinearity and orthogonality to solve problems involving vectors.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Collinear points and orthogonal vectors: Outcome Check

Outcome check for Collinear points and orthogonal vectors: State the conditions and properties of collinearity and orthogonality.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Enlargement and similarity in 2D: Lesson Opener

Learning focus: Enlargement and similarity in 2D

Country link: a Rwandan school, market, farming, building, transport, or data problem for Enlargement and similarity in 2D.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve problems regarding shape enlargement and similarities in 2D
- identify similar shapes
- list properties of enlargement and similarities
- determine the linear scale factor of an enlargement
- find the centre of an

Inquiry questions:

- How can enlargement and similarity in 2d help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Enlargement and similarity in 2D: Learn

Enlargement and similarity in 2D teaches properties of two-dimensional shapes.

Core idea:

- Shapes can be classified by sides, corners, lines of symmetry, and angles.
- A labelled sketch helps you compare shapes accurately.
- Use correct names such as triangle, rectangle, square, and circle.

Key words:

- Enlargement and similarity in 2D
- Enlargement
- similarity
- method
- model
- unit
- answer
- check

Enlargement and similarity in 2D: Worked Example

Worked example for Enlargement and similarity in 2D: Classify a shape with 4 equal sides and 4 right angles.

1. Four sides means it is a quadrilateral.
2. Equal sides and right angles match a square.
3. Check the name and properties.

Answer: The shape is a square.

Enlargement and similarity in 2D: Vocabulary And Study Skill

Use these words and study moves before you practise Enlargement and similarity in 2D. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Enlargement and similarity in 2D
- Enlargement
- similarity
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Enlargement and similarity in 2D without a clear example, method, or evidence.

Enlargement and similarity in 2D: Guided Activity

Guided activity for Enlargement and similarity in 2D:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Enlargement and similarity in 2D: Practice And Correction

Practice for Enlargement and similarity in 2D:

Main outcome: solve problems regarding shape enlargement and similarities in 2D.

1. Draw a triangle, rectangle, and square.
2. Write two properties for each shape.
3. Circle the shape that has four equal sides and four right angles.

Answer check:

Answer: the square has four equal sides and four right angles.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to enlargement and similarity in 2d and solve it.

Enlargement and similarity in 2D: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Enlargement and similarity in 2D.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Enlargement and similarity in 2D: Outcome Check

Outcome check for Enlargement and similarity in 2D: Solve problems regarding shape enlargement and similarities in 2D.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Enlargement and similarity in 2D: Outcome Check

Outcome check for Enlargement and similarity in 2D: Identify similar shapes.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Inverse and composite transformations in 2D: Lesson Opener

Learning focus: Inverse and composite transformations in 2D

Country link: a Rwandan school, market, farming, building, transport, or data problem for Inverse and composite transformations in 2D.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve problems involving the inverse and composite transformations of shapes
- state and explain properties of composite and inverse transformations in 2D
- identify type of transformation used in given drawings in 2D
- show an image of an object from different transformed shapes in 2D
- construct an image of an object under composite and inverse transformation in 2D
- solve problems involving inverse and composite transformations in 2D

Inquiry questions:

- How can inverse and composite transformations in 2D help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Inverse and composite transformations in 2D: Learn

Inverse and composite transformations in 2D teaches properties of two-dimensional shapes.

Core idea:

- Shapes can be classified by sides, corners, lines of symmetry, and angles.
- A labelled sketch helps you compare shapes accurately.
- Use correct names such as triangle, rectangle, square, and circle.

Key words:

- Inverse and composite transformations in 2D
- Inverse
- composite
- transformations
- method
- model
- unit
- answer

Inverse and composite transformations in 2D: Worked Example

Worked example for Inverse and composite transformations in 2D: Classify a shape with 4 equal sides and 4 right angles.

1. Four sides means it is a quadrilateral.
2. Equal sides and right angles match a square.
3. Check the name and properties.

Answer: The shape is a square.

Inverse and composite transformations in 2D: Vocabulary And Study Skill

Use these words and study moves before you practise Inverse and composite transformations in 2D. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Inverse and composite transformations in 2D
- Inverse
- composite
- transformations
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Inverse and composite transformations in 2D without a clear example, method, or evidence.

Inverse and composite transformations in 2D: Guided Activity

Guided activity for Inverse and composite transformations in 2D:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Inverse and composite transformations in 2D: Practice And Correction

Practice for Inverse and composite transformations in 2D:

Main outcome: solve problems involving the inverse and composite transformations of shapes.

1. Draw a triangle, rectangle, and square.
2. Write two properties for each shape.
3. Circle the shape that has four equal sides and four right angles.

Answer check:

Answer: the square has four equal sides and four right angles.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to inverse and composite transformations in 2d and solve it.

Inverse and composite transformations in 2D: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Inverse and composite transformations in 2D.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Inverse and composite transformations in 2D: Outcome Check

Outcome check for Inverse and composite transformations in 2D: Solve problems involving the inverse and composite transformations of shpaes.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Inverse and composite transformations in 2D: Outcome Check

Outcome check for Inverse and composite transformations in 2D: State and explain properties of composite and inverse transformations in 2D.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics (bivariate data): Lesson Opener

Learning focus: Statistics (bivariate data)

Country link: a Rwandan school, market, farming, building, transport, or data problem for Statistics (bivariate data).

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- collect, represent, and interpret bivariate data
- define bivariate data
- make a frequency distribution table of collected bivariate data
- interpret scatter diagrams
- identify type of correlation on a scatter diagram
- draw scatter diagram for bivariate data and indicate the type of correlation

Inquiry questions:

- How can statistics (bivariate data) help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics (bivariate data): Learn

Statistics (bivariate data) teaches how to collect, organize, read, and explain information.

Core idea:

- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:

- Statistics (bivariate data)
- Statistics
- bivariate
- data
- method
- model

Statistics (bivariate data): Worked Example

Worked example for Statistics (bivariate data): A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Statistics (bivariate data): Vocabulary And Study Skill

Use these words and study moves before you practise Statistics (bivariate data). Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Statistics (bivariate data)
- Statistics
- bivariate
- data
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Statistics (bivariate data) without a clear example, method, or evidence.

Statistics (bivariate data): Guided Activity

Guided activity for Statistics (bivariate data):

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Statistics (bivariate data): Practice And Correction

Practice for Statistics (bivariate data):

Main outcome: collect, represent, and interpret bivariate data.

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:

Answer check: totals in the tally table and bar chart should match.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to statistics (bivariate data) and solve it.

Statistics (bivariate data): Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Statistics (bivariate data).
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics (bivariate data): Outcome Check

Outcome check for Statistics (bivariate data): Collect, represent, and interpret bivariate data.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics (bivariate data): Outcome Check

Outcome check for Statistics (bivariate data): Define bivariate data.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Problem sets: Review Clinic 1

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Problem sets that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Number bases: Review Clinic 2

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Number bases that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Algebraic fractions: Review Clinic 3

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Algebraic fractions that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Simultaneous linear equations and inequalities: Review Clinic 4

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Simultaneous linear equations and inequalities that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Mathematics.

- Problem sets: Explain this term using an example from the book, your school, or your community.
- Number bases: Explain this term using an example from the book, your school, or your community.
- Algebraic fractions: Explain this term using an example from the book, your school, or your community.
- Simultaneous linear equations and inequalities: Explain this term using an example from the book, your school, or your community.
- Quadratic equations: Explain this term using an example from the book, your school, or your community.
- Linear and quadratic functions: Explain this term using an example from the book, your school, or your community.
- Compound interest, reverse percentage, and compound proportional change: Explain this term using an example from the book, your school, or your community.
- Right angle triangles: Explain this term using an example from the book, your school, or your community.
- Circle theorem: Explain this term using an example from the book, your school, or your community.
- Collinear points and orthogonal vectors: Explain this term using an example from the book, your school, or your community.
- Enlargement and similarity in 2D: Explain this term using an example from the book, your school, or your community.
- Inverse and composite transformations in 2D: Explain this term using an example from the book, your school, or your community.
- Statistics (bivariate data): Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Mathematics that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.